



CREDIT CARD ANALYSIS

SQL PORTFOLIO PROJECT

HOW IT WORKS

- ✓ **MEASURE**
1234 saction Analysis
- ✓ **SEGMENT**
Business Questions
- ✓ **GROW**
Actionable Insights

- ✓ Top Cities by Spend
- ✓ Customer Spending Behavior
- ✓ MoM Growth Analysis
- ✓ Business KPIs
- ✓ SQL Window Functions



MS SQL Server

Saideep Pallela

Data Analyst | SQL

 /saideeppallela



TRANSACTIONAL SQL



ANALYTICS



BUSINESS INSIGHTS



HIRABLE PROJECT

Measure. Segment. Grow.



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Credit Card Transactions Analysis (SQL Project)

Author :- Saideep Pallela

Data Analyst (SQL • Excel • Power BI)

Tools & Technologies: Microsoft SQL Server (T-SQL)

Dataset: Credit Card Spending Behaviour in India (Kaggle)

Objective:

This project analyses real-world credit card transaction data to identify spending patterns across cities, card types, and expense categories.

The focus is on transforming raw transactional data into meaningful business insights that highlight customer behaviour, usage trends, and key drivers of credit card spending.

Dataset Overview

Table Name :- credit_card_transactions

Column Name	Description
transaction_id	Unique identifier for each transaction
transaction_date	Date on which the transaction occurred
city	City where the transaction was recorded
card_type	Type of credit card used (e.g., Gold, Silver, Platinum, Signature)
exp_type	Category of expense (e.g., Bills, Fuel, Travel, Entertainment)
gender	Gender of the cardholder (Male/Female)
amount	Monetary value of the transaction

Data Validation & Quality Checks

Before starting the analysis, I performed basic data validation to understand the structure and ensure the dataset is reliable.

Sample Data Check

```
SELECT TOP 10 *  
FROM credit_card_transactions;
```

	transaction_id	city	transaction_date	card_type	exp_type	gender	amount
1	1	Delhi	2014-10-29	Gold	Bills	F	82475.00
2	2	Greater Mumbai	2014-08-22	Platinum	Bills	F	32555.00
3	3	Bengaluru	2014-08-27	Silver	Bills	F	101738.00
4	4	Greater Mumbai	2014-04-12	Signature	Bills	F	123424.00
5	5	Bengaluru	2015-05-05	Gold	Bills	F	171574.00
6	6	Delhi	2014-09-08	Silver	Bills	F	100036.00
7	7	Delhi	2015-02-24	Gold	Bills	F	143250.00
8	8	Greater Mumbai	2014-06-26	Platinum	Bills	F	150980.00
9	9	Delhi	2014-03-28	Silver	Bills	F	192247.00
10	10	Delhi	2014-09-01	Platinum	Bills	F	67932.00



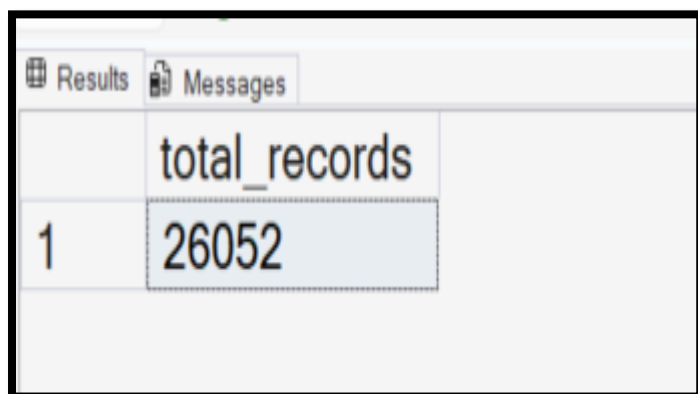
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Sample Data Check Insight:

The dataset contains transaction-level records with a mix of categorical and numerical columns, suitable for analysis.

Record Count Check

```
SELECT COUNT(*) AS total_records  
FROM credit_card_transactions;
```



A screenshot of a database query result window. It has two tabs: 'Results' and 'Messages'. The 'Results' tab is active, showing a table with one column named 'total_records' and one row with the value '26052'.

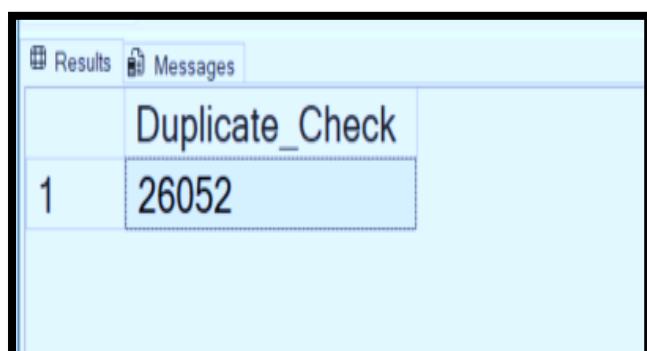
	total_records
1	26052

Record Count Check Insight:

The dataset contains 26052 records with a mix of categorical and numerical columns, suitable for analysis.

Duplicate Check

```
SELECT COUNT(DISTINCT transaction_id) as Duplicate_check  
FROM credit_card_transactions;
```



A screenshot of a database query result window. It has two tabs: 'Results' and 'Messages'. The 'Results' tab is active, showing a table with one column named 'Duplicate_Check' and one row with the value '26052'.

	Duplicate_Check
1	26052

Duplicate Check Insight:

No duplicate transaction IDs were found, indicating that the data is consistent.



NULL Value Check

```
SELECT *  
FROM credit_card_transactions  
WHERE amount IS NULL;
```

transaction_id	city	transaction_date	card_type	exp_type	gender	amount
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NULL Value Check Insight:

No missing values were observed .

Exploratory Analysis

After validating the data, I performed basic analysis to understand overall spending behavior.

Total Spend

```
SELECT SUM(amount) AS total_spend  
FROM credit_card_transactions;
```

	total_spend
1	4074833373.00

Total Spend Insight:

This gives an overall view of the total transaction value in the dataset.

Spend by Card Type

```
SELECT card_type, SUM(amount) AS total_spend  
FROM credit_card_transactions  
GROUP BY card_type;
```

	card_type	total_spend
1	Gold	984539536.00
2	Signature	1013041105.00
3	Platinum	1007639019.00
4	Silver	1069613713.00

Spend by Card Type Insight:

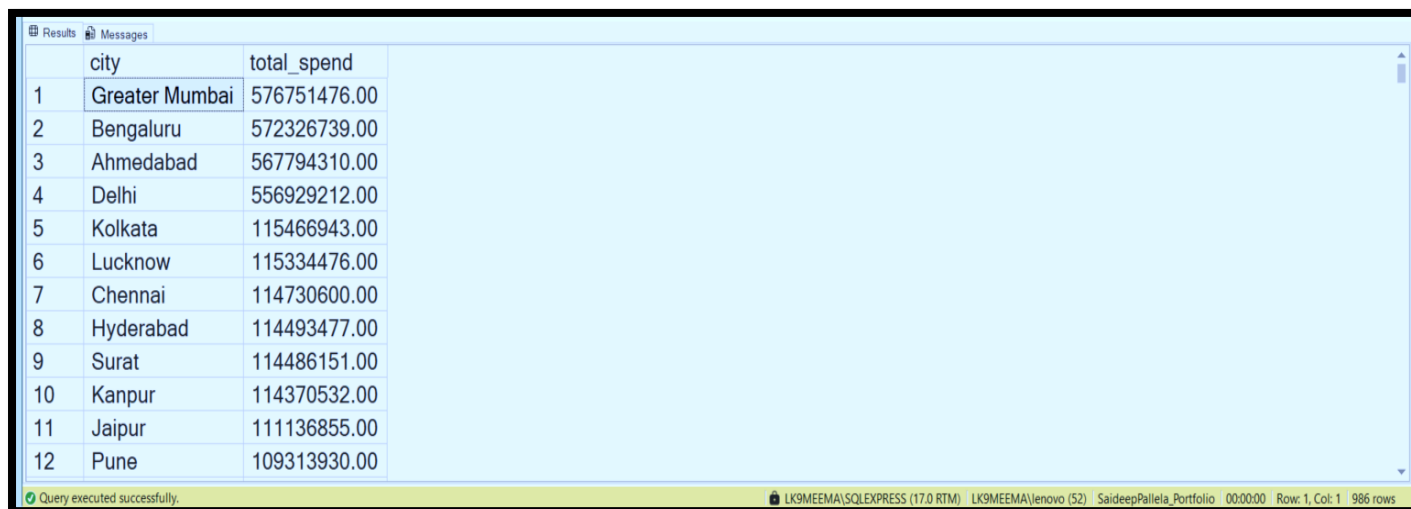
Some card types contribute more to total spending, indicating differences in customer usage patterns.



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Spend by City

```
SELECT city, SUM(amount) AS total_spend
FROM credit_card_transactions
GROUP BY city
ORDER BY total_spend DESC;
```



The screenshot shows a SQL Server Results window with a table containing 12 rows of data. The table has two columns: 'city' and 'total_spend'. The data is sorted in descending order of total spend. The status bar at the bottom indicates 'Query executed successfully.' and 'Row: 1, Col: 1 986 rows'.

	city	total_spend
1	Greater Mumbai	576751476.00
2	Bengaluru	572326739.00
3	Ahmedabad	567794310.00
4	Delhi	556929212.00
5	Kolkata	115466943.00
6	Lucknow	115334476.00
7	Chennai	114730600.00
8	Hyderabad	114493477.00
9	Surat	114486151.00
10	Kanpur	114370532.00
11	Jaipur	111136855.00
12	Pune	109313930.00

Spend by City Insight:

Spending is higher in major cities, showing that urban areas contribute more to credit card usage.

Key Initial Insights:

Credit card spending is higher in major cities compared to smaller locations.

Certain card types show higher usage, indicating different customer segments.

The dataset is clean and suitable for further analysis using advanced SQL concepts.

Business Problems Solving using T-SQL (Microsoft SQL Server)

After understanding the dataset and identifying initial patterns, I proceeded to solve specific business-driven questions to extract deeper insights from the data.

The following analysis focuses on real-world scenarios such as identifying high-performing cities, understanding spending distribution, and deriving actionable business insights using T-SQL.

1. Top 5 Cities by Credit Card Spend Contribution

Business Question

Identify the top 5 cities with the highest credit card spending and calculate their percentage contribution to the overall spend.

Approach

To solve this problem, I calculated the total spend for each city and the overall spend across all transactions.

Then, I derived the percentage contribution of each city and ranked them to extract the top 5 cities.

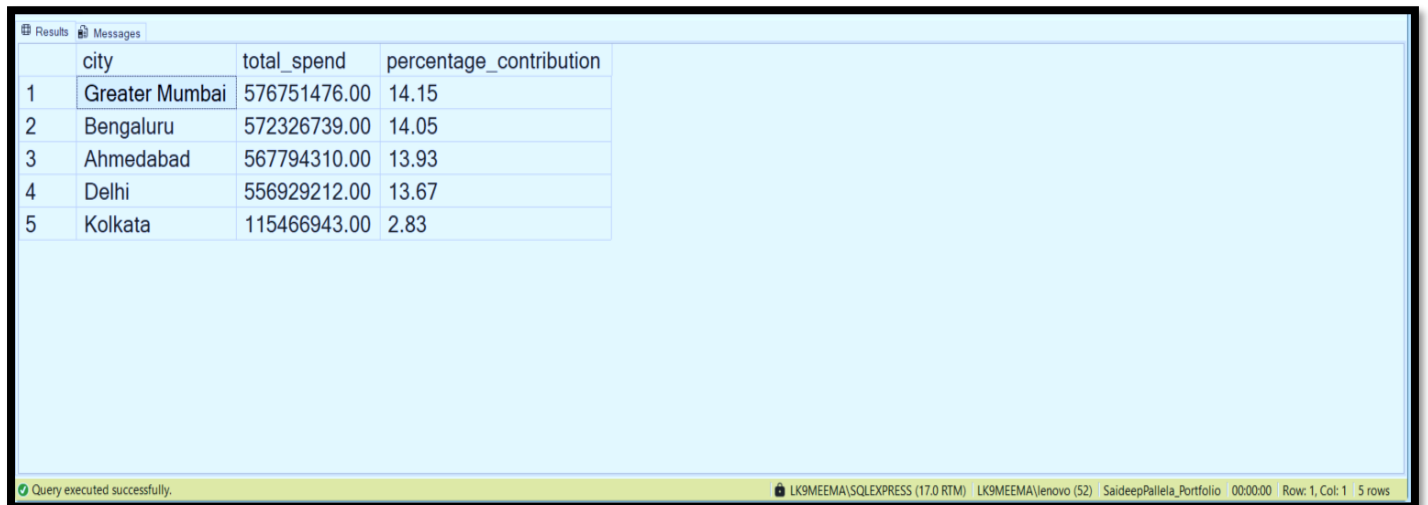


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SQL Query

```
WITH city_spend AS (  
    SELECT  
        city,  
        SUM(amount) AS total_spend  
    FROM credit_card_transactions  
    GROUP BY city  
)  
overall_spend AS (  
    SELECT SUM(amount) AS total_amount  
    FROM credit_card_transactions  
)  
SELECT TOP 5  
    cs.city,  
    cs.total_spend,  
    CAST(ROUND(cs.total_spend * 100.0 / os.total_amount, 2) AS  
DECIMAL(5,2)) AS percentage_contribution  
FROM city_spend AS cs  
CROSS JOIN overall_spend AS os  
ORDER BY cs.total_spend DESC;
```

Output (Query Result)



	city	total_spend	percentage_contribution
1	Greater Mumbai	576751476.00	14.15
2	Bengaluru	572326739.00	14.05
3	Ahmedabad	567794310.00	13.93
4	Delhi	556929212.00	13.67
5	Kolkata	115466943.00	2.83

Query executed successfully. LK9MEEMA\SQLEXPRESS (17.0 RTM) LK9MEEMA\lenovo (52) SaideepPallela_Portfolio 00:00:00 Row: 1, Col: 1 5 rows

Key Insight:

The top 4 cities contribute almost equally (around 13–14% each), while there is a sharp drop in contribution from the 5th city (Kolkata), indicating an uneven distribution of credit card spending.

Business Interpretation

This pattern suggests that credit card usage is highly concentrated in a few major cities. Businesses can focus on these high-performing regions for targeted marketing, premium offerings, and customer retention strategies, while also identifying opportunities to grow usage in lower-performing cities.



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2. Highest Spend Month for Each Card Type

Business Question

Identify the month with the highest total spending for each card type.

Approach

To solve this, I first aggregated the total spend for each card type on a monthly level. Then, I used a window function to rank the months based on total spending within each card type. Finally, I filtered the top-ranked month for each card type.

SQL Query

```
WITH monthly_spend AS (  
    SELECT  
        card_type,  
        FORMAT(transaction_date, 'yyyy-MM') AS year_month,  
        SUM(amount) AS total_spend  
    FROM credit_card_transactions  
    GROUP BY  
        card_type,  
        FORMAT(transaction_date, 'yyyy-MM')  
)  
ranked_spend AS (  
    SELECT *,  
        RANK() OVER (PARTITION BY card_type ORDER BY total_spend DESC)  
    AS rank_num  
    FROM monthly_spend  
)  
SELECT  
    card_type,  
    year_month,  
    total_spend  
FROM ranked_spend  
WHERE rank_num = 1;
```

Output (Query Result)



	card_type	year_month	total_spend
1	Gold	2015-01	55455064.00
2	Platinum	2014-08	57936507.00
3	Signature	2013-12	58799522.00
4	Silver	2015-03	59723549.00

Query executed successfully.



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Key Insight

Each card type has a specific peak month where spending is highest, indicating seasonal or behavioural patterns in customer transactions.

Business Interpretation

Understanding peak spending months helps businesses plan targeted campaigns, promotional offers, and credit strategies aligned with customer behavior for each card segment.

3. First Transaction When Cumulative Spend Reaches 1,000,000

Business Question

Identify the transaction details for each card type at the point where cumulative spending first reaches 1,000,000.

Approach

To solve this problem, I calculated the cumulative (running) total of transaction amounts for each card type using a window function.

Then, I filtered the records where the cumulative spend crossed 1,000,000.

Finally, I selected the first such transaction for each card type.

SQL Query

```
WITH cumulative_spend AS (
    SELECT
        *,
        SUM(amount) OVER (
            PARTITION BY card_type
            ORDER BY transaction_date, transaction_id
        ) AS running_total
    FROM credit_card_transactions
)
SELECT *
FROM (
    SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY card_type
            ORDER BY running_total
        ) AS rn
    FROM cumulative_spend
    WHERE running_total >= 1000000
) t
WHERE rn = 1;
```



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Output (Query Result)

	transaction_id	city	transaction_date	card_type	exp_type	gender	amount	running_total	rn
1	1522	Delhi	2013-10-04	Gold	Food	M	281924.00	1272624.00	1
2	191	Ahmedabad	2013-10-05	Platinum	Bills	F	612572.00	1537482.00	1
3	73	Delhi	2013-10-04	Signature	Bills	F	550782.00	1285819.00	1
4	7565	Bengaluru	2013-10-04	Silver	Food	F	205179.00	1115582.00	1

Query executed successfully.

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Key Insight

Each card type reaches the 1,000,000 cumulative spend milestone at different points in time, reflecting differences in transaction frequency and spending behavior.

Business Interpretation

This analysis helps identify how quickly different customer segments generate revenue. Faster accumulation indicates higher engagement and spending activity, which can guide targeted strategies for high-value customers.

4. City with Lowest Percentage Spend for Gold Card

Business Question

Identify the city that contributes the lowest percentage of total spending for the Gold card category.

Approach

To solve this, I first calculated the total gold card spending for each city.

Then, I computed the overall gold card spending across all cities.

Finally, I calculated the percentage contribution for each city and selected the city with the lowest contribution.

SQL Query

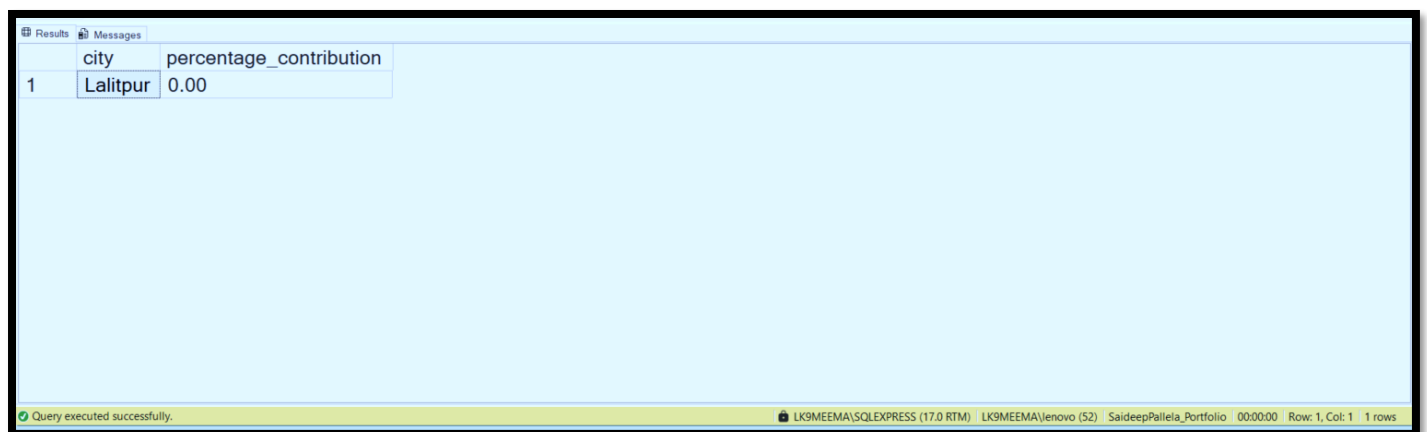
```
WITH gold_city_spend AS (  
    SELECT  
        city,  
        SUM(amount) AS gold_spend  
    FROM credit_card_transactions  
    WHERE card_type = 'Gold'  
    GROUP BY city  
)
```



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```
total_gold_spend AS (
    SELECT SUM(amount) AS total_spend
    FROM credit_card_transactions
    WHERE card_type = 'Gold'
)
SELECT TOP 1
    gcs.city,
    CAST(ROUND(gcs.gold_spend * 100.0 / tgs.total_spend, 2) AS
DECIMAL(5,2)) AS percentage_contribution
FROM gold_city_spend AS gcs
CROSS JOIN total_gold_spend AS tgs
ORDER BY percentage_contribution ASC;
```

Output (Query Result)



	city	percentage_contribution
1	Lalitpur	0.00

Key Insight

Some cities contribute very little to Gold card spending, indicating low adoption or usage of premium cards in those regions.

Business Interpretation

This insight helps identify underperforming regions for premium card usage. Businesses can target these cities with awareness campaigns or tailored offers to increase adoption of higher-value card products.

5. Highest and Lowest Expense Type by City

Business Question

For each city, identify the expense category with the highest and lowest total spending

Approach

To solve this, I first calculated total spending for each expense type within each city.

Then, I used window functions to rank expense categories based on total spend in both ascending and descending order.

Finally, I extracted the highest and lowest expense types for each city.

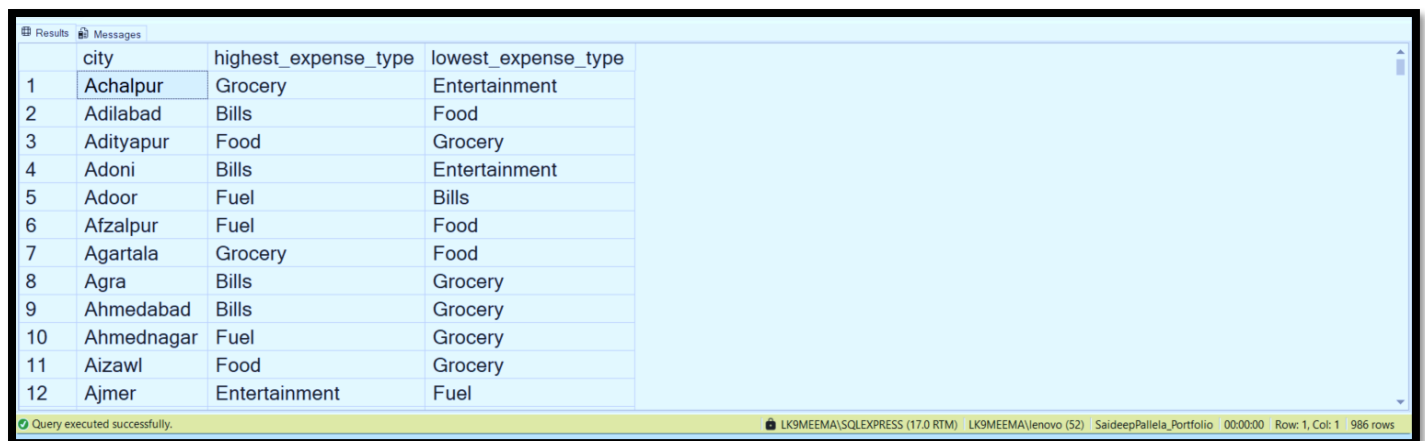


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SQL Query

```
WITH city_expense AS (  
    SELECT  
        city,  
        exp_type,  
        SUM(amount) AS total_amount  
    FROM credit_card_transactions  
    GROUP BY city, exp_type  
)  
ranked_expense AS (  
    SELECT *,  
        ROW_NUMBER() OVER (PARTITION BY city ORDER BY total_amount  
DESC) AS rn_high,  
        ROW_NUMBER() OVER (PARTITION BY city ORDER BY total_amount ASC)  
AS rn_low  
    FROM city_expense  
)  
SELECT  
    city,  
    MAX(CASE WHEN rn_high = 1 THEN exp_type END) AS highest_expense_type,  
    MAX(CASE WHEN rn_low = 1 THEN exp_type END) AS lowest_expense_type  
FROM ranked_expense  
GROUP BY city;
```

Output (Query Result)



	city	highest_expense_type	lowest_expense_type
1	Achalpur	Grocery	Entertainment
2	Adilabad	Bills	Food
3	Adityapur	Food	Grocery
4	Adoni	Bills	Entertainment
5	Adoor	Fuel	Bills
6	Afzalpur	Fuel	Food
7	Agartala	Grocery	Food
8	Agra	Bills	Grocery
9	Ahmedabad	Bills	Grocery
10	Ahmednagar	Fuel	Grocery
11	Aizawl	Food	Grocery
12	Ajmer	Entertainment	Fuel

Query executed successfully. LK9MEEMA\SQL EXPRESS (17.0 RTM) LK9MEEMA\lenovo (52) SaideepPallela_Portfolio 00:00:00 Row: 1, Col: 1 986 rows

Key Insight

Spending patterns vary significantly across cities, with certain expense categories dominating in each location.

Business Interpretation

This analysis helps businesses understand local spending behavior, enabling them to tailor offers and services based on city-specific preferences.



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6. Female Contribution to Spending by Expense Type

Business Question

Determine the percentage contribution of female customers to total spending across different expense categories.

Approach

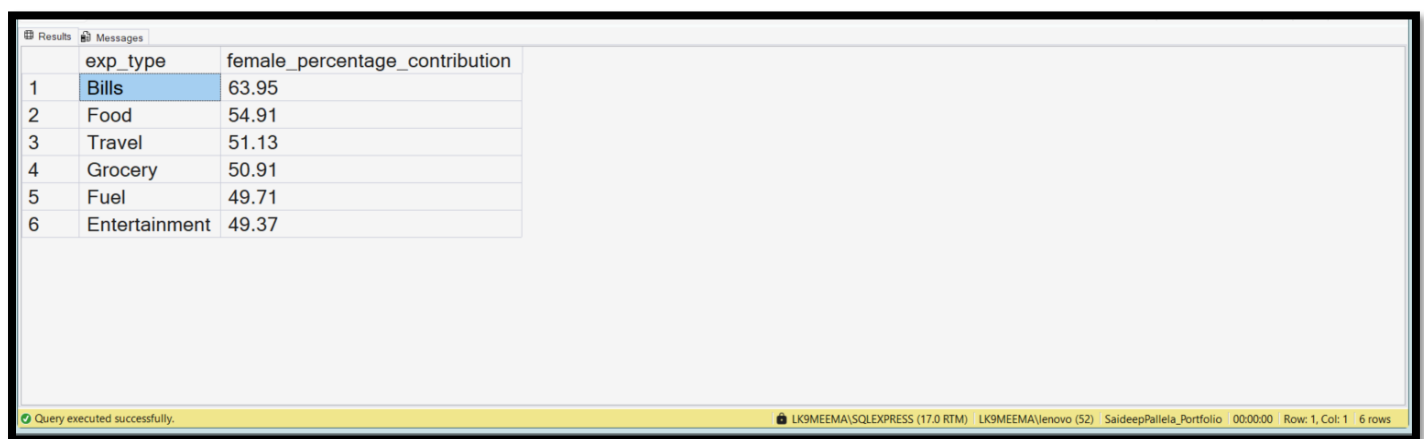
To solve this, I calculated the total spend contributed by female customers for each expense type using conditional aggregation.

Then, I divided it by the total spend for that expense type to get the percentage contribution.

SQL Query

```
SELECT
    exp_type,
    CAST(
        ROUND(
            SUM(CASE WHEN gender = 'F' THEN amount ELSE 0 END) * 100.0
            / SUM(amount),
            2)
        AS DECIMAL(5,2)) AS female_percentage_contribution
FROM credit_card_transactions
GROUP BY exp_type
ORDER BY female_percentage_contribution DESC;
```

Output (Query Result)



	exp_type	female_percentage_contribution
1	Bills	63.95
2	Food	54.91
3	Travel	51.13
4	Grocery	50.91
5	Fuel	49.71
6	Entertainment	49.37

Key Insight

Female contribution to spending varies across expense categories, with certain categories showing higher engagement from female customers.

Business Interpretation

This insight can help businesses design targeted marketing strategies and personalized offers for female customers in specific categories where their engagement is higher.



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7. Highest Month-over-Month Growth (Jan 2014)

Business Question

Identify which combination of card type and expense category experienced the highest month-over-month growth in January 2014.

Approach

To solve this, I first calculated the monthly spend for each card type and expense category.

Then, I used a window function (LAG) to get the previous months spend.

Finally, I computed the month-over-month growth and identified the highest growth for January 2014.

SQL Query

```
WITH monthly_spend AS (  
    SELECT  
        card_type,  
        exp_type,  
        FORMAT(transaction_date, 'yyyy-MM') AS year_month,  
        SUM(amount) AS total_spend  
    FROM credit_card_transactions  
    GROUP BY  
        card_type,  
        exp_type,  
        FORMAT(transaction_date, 'yyyy-MM')  
) ,  
mom_calculation AS (  
    SELECT *,  
        LAG(total_spend) OVER (  
            PARTITION BY card_type, exp_type  
            ORDER BY year_month  
        ) AS prev_month_spend  
    FROM monthly_spend  
)  
SELECT TOP 1  
    card_type,  
    exp_type,  
    year_month,  
    total_spend,  
    prev_month_spend,  
    (total_spend - prev_month_spend) AS mom_growth  
FROM mom_calculation  
WHERE year_month = '2014-01'  
    AND prev_month_spend IS NOT NULL  
ORDER BY mom_growth DESC.
```



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Output (Query Result)

	card_type	exp_type	year_month	total_spend	prev_month_spend	mom_growth
1	Platinum	Grocery	2014-01	12256343.00	7757562.00	4498781.00

Query executed successfully.

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Key Insight

Certain card and expense combinations show significant spikes in spending during specific months, indicating strong seasonal or behavioral trends.

Business Interpretation

This insight helps businesses identify high-growth segments and optimize promotions or offers during peak periods to maximize revenue.

8. City with Highest Spend per Transaction on Weekends

Business Question

Identify the city with the highest average spend per transaction during weekends.

Approach

To solve this, I filtered transactions that occurred on weekends (Saturday and Sunday). Then, I calculated the total spend and divided it by the number of transactions for each city to derive the average spend per transaction. Finally, I ranked the cities to identify the highest value.

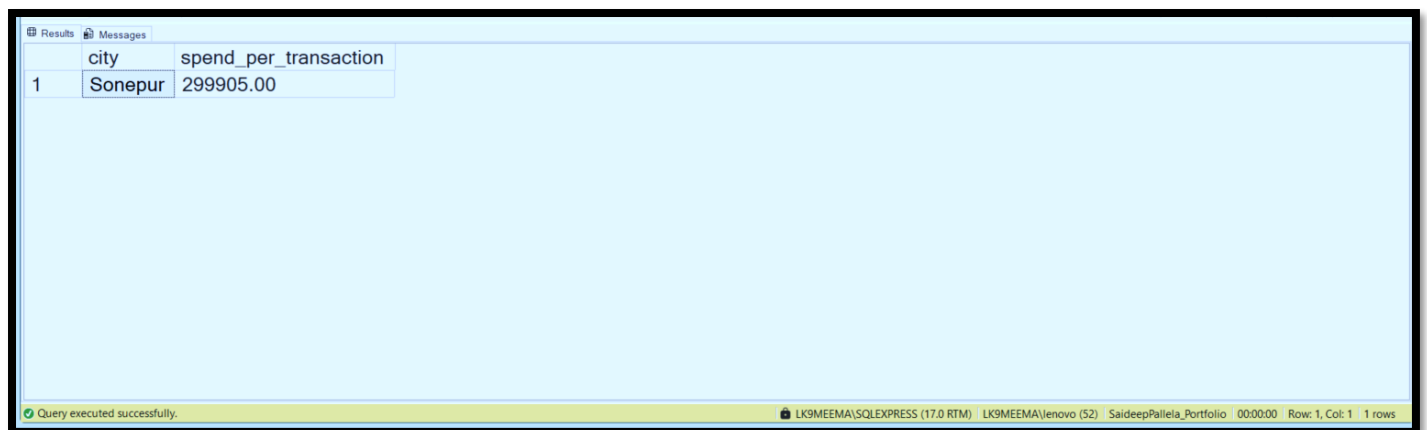
SQL Query

```
SELECT TOP 1
    city,
    CAST(
        ROUND(
            SUM(amount) * 1.0 / COUNT(*),
            2)
        AS DECIMAL(10,2)) AS spend_per_transaction
FROM credit_card_transactions
WHERE DATENAME(WEEKDAY, transaction_date) IN ('Saturday', 'Sunday')
GROUP BY city
ORDER BY spend_per_transaction DESC;
```



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Output (Query Result)



	city	spend_per_transaction
1	Sonepur	299905.00

Query executed successfully. LK9MEEMA\SQLEXPRESS (17.0 RTM) LK9MEEMA\lenovo (52) SaideepPallela_Portfolio 00:00:00 Row: 1, Col: 1 1 rows

Key Insight

Certain cities show higher average spend per transaction during weekends, indicating more premium or high-value purchases.

Business Interpretation

This insight helps businesses identify locations where customers spend more per visit on weekends, enabling targeted promotions, pricing strategies, and premium offerings.

9. City with Fastest Growth to 500 Transactions

Business Question

Identify the city that took the least number of days to reach its 500th transaction after the first transaction.

Approach

To solve this, I first assigned a row number to each transaction within each city based on transaction date.

Then, I identified the first and 500th transactions for each city.

Finally, I calculated the number of days between these two transactions and selected the city with the minimum duration.

SQL Query

```
WITH ranked_transactions AS (  
    SELECT  
        city,  
        transaction_date,  
        ROW_NUMBER() OVER (  
            PARTITION BY city  
            ORDER BY transaction_date, transaction_id  
        ) AS rn
```



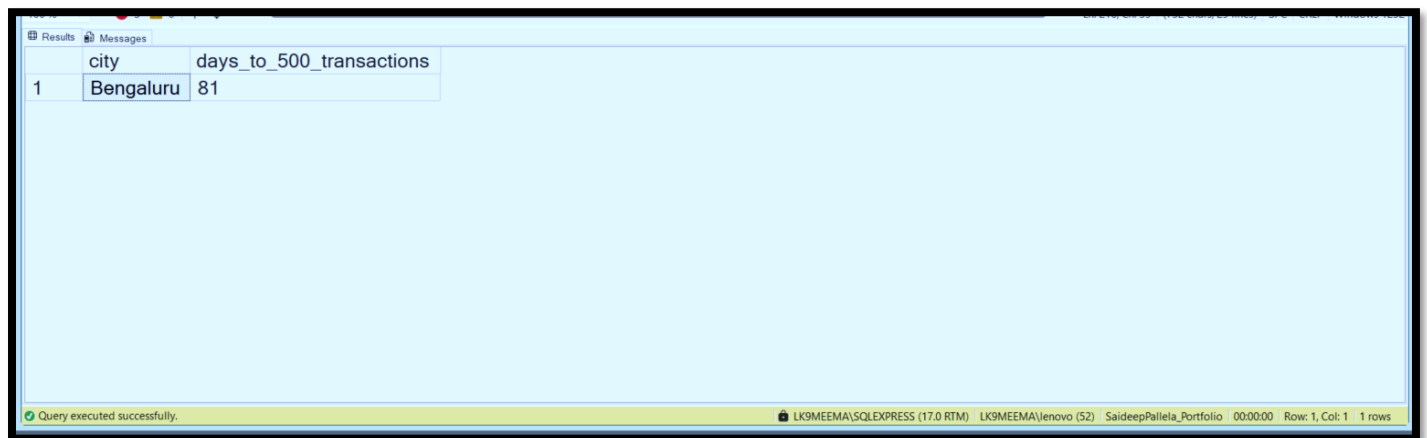
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```

FROM credit_card_transactions
),
first_500_transactions AS (
    SELECT
        city,
        MIN(transaction_date) AS first_transaction_date,
        MAX(transaction_date) AS transaction_500_date
    FROM ranked_transactions
    WHERE rn IN (1, 500)
    GROUP BY city
    HAVING COUNT(*) = 2
)
SELECT TOP 1
    city,
    DATEDIFF(
        DAY,
        first_transaction_date,
        transaction_500_date
    ) AS days_to_500_transactions
FROM first_500_transactions
ORDER BY days_to_500_transactions ASC;

```

Output (Query Result)



	city	days_to_500_transactions
1	Bengaluru	81

Query executed successfully. LK9MEEMA\SQLEXPRESS (17.0 RTM) LK9MEEMA\lenovo (52) SaideepPallela_Portfolio 00:00:00 Row: 1, Col: 1 1 rows

Key Insight

Some cities reach transaction milestones much faster than others, indicating higher transaction frequency and customer activity.

Business Interpretation

This insight helps identify high-growth cities where customer engagement is strong. Businesses can prioritize these locations for expansion, marketing, and customer retention strategies.



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